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ORIGINAL ARTICLE

Pathological brain plasticity and cognition in the offspring of males subjected to postnatal traumatic stress

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Traumatic stress in early-life increases the risk for cognitive and neuropsychiatric disorders later in life. Such early stress can also impact the progeny even if not directly exposed, likely through epigenetic mechanisms. Here, we report in mice that the offspring of males subjected to postnatal traumatic stress have decreased gene expression in molecular pathways necessary for neuronal signaling, and altered synaptic plasticity when adult. Long-term potentiation is abolished and long-term depression is enhanced in the hippocampus, and these defects are associated with impaired long-term memory in both the exposed fathers and their offspring. The brain-specific gamma isoform of protein kinase C (*Prkcc*) is one of the affected signaling components in the hippocampus. Its expression is reduced in the offspring, and DNA methylation at its promoter is altered both in the hippocampus of the offspring and the sperm of fathers. These results suggest that postnatal traumatic stress in males can affect brain plasticity and cognitive functions in the adult progeny, possibly through epigenetic alterations in the male germline.

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INTRODUCTION

Early life stress resulting from emotional and physical neglect or abuse in childhood is a major risk factor for the development of psychiatric conditions such as major depression, bipolar disorder, schizophrenia and post-traumatic stress disorder.[1,2] Considering that reported cases of child abuse or neglect are around 10% in several European countries and the United States,[3–6] the long-term consequences of such trauma place a heavy burden on society and on healthcare systems worldwide.[2] Recent studies in mice have demonstrated that exposure of male pups to traumatic stress involving repeated episodes of unpredictable maternal separation combined with unpredictable maternal stress (MSUS), leads to depressive-like behaviors, altered risk assessment and impaired social interactions in adulthood across several generations.[7–10] This suggests that paternal trauma is a risk factor for the development of behavioral disorders in the progeny.[11,12] In addition to affective and emotional disorders, cognitive dysfunctions are also common to many conditions induced by stress,[13–16] in part because the hippocampus, a brain region critical for learning and memory formation, is an important component of stress response pathways.[17,18] Animal models have established that early trauma can have lifelong negative consequences on cognitive performance and synaptic plasticity in the hippocampus of exposed animals,[19–21] but the impact on the offspring has not yet been carefully assessed.

Using the MSUS mouse model, we conducted an unbiased, genome-wide analysis of gene expression in the adult hippocampus following postnatal traumatic stress, and examined whether gene networks are affected in the offspring of stressed males. Here, we show that the offspring of males exposed to MSUS have widespread alterations in gene expression in the hippocampus, specifically in molecular networks implicated in synaptic plasticity. The animals also have a dramatic shift in functional synaptic plasticity, in particular abolished long-term potentiation (LTP) and enhanced long-term depression (LTD). This shift is accompanied by impaired hippocampus-dependent long-term memory. We identify the brain-specific gamma subunit of protein kinase C (*Prkcc*), a gene implicated in synaptic plasticity and memory performance,[22,23] as a potential molecular target. *Prkcc* expression is decreased in the hippocampus of the offspring, and DNA methylation is reduced at a transcription factor-binding site of its promoter region both in the brain of the offspring and the sperm of fathers.

MATERIALS AND METHODS

Animals

Mice were maintained in a temperature- and humidity-controlled facility on a 12 h reversed light–dark cycle with food and water *ad libitum*. A total of 323 mice (285 males and 38 females) were used for this study. All procedures were carried out in accordance to Swiss cantonal regulations for animal experimentation under license numbers 210/2008 and 55/2012.

MSUS, breeding paradigm and cross-fostering

MSUS was conducted as previously described.[8] Briefly, C57Bl/6JRj male and female mice (F0, Elevage Janvier, Le Genest Saint Isle, France) were bred (one-to-one pairing for 7 days). Gestating females were singly housed and subjected to MSUS starting 1 day after delivery, or were allowed to raise their pups normally (controls). Only dams giving birth within a week of each other and with litters of more than four pups were used. For MSUS, pups (F1) were separated from their mother for 3 h per day at unpredictable times from postnatal day 1–14. During separation, mothers were randomly exposed to restraint stress (20 min) or forced swim stress (5 min) unpredictably. At postnatal day 21, pups were weaned and placed in standard cages (3–5 mice per cage, each from a different litter to avoid

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litter effects). When adult, F1 males subjected to MSUS and controls were bred to naïve wild-type females to produce F2 offspring, and F2 offspring were bred to naïve wild-type females when adult to produce F3 offspring. F2 and F3 mice were reared in normal conditions and were never exposed to early-life stress (see Supplementary Figure S1 for experimental design). F1 males and the derived F2 and F3 animals constitute one batch, and three independent batches of F1, F2 and F3 animals were generated for this study.

Cross-fostering. For cross-fostering, MSUS and control litters (selected for same litter size and birthdate) were switched to a foster mother from the opposite experimental group (MSUS pups to control dams and control pups to MSUS dams) within 48 h after birth as described previously.[24]

Gene expression analyses

Adult mice were killed by cervical dislocation either 45 min (acute stress condition) or 2 weeks (baseline resting condition) after exposure to forced swim stress (6 min, 18 °C water) (conducted in two independent experiments using mice from different breeding batches, see Supplementary Figure S2A, B for experimental design). The brain was removed and the hippocampus rapidly dissected on ice and stored at −80 °C until further processing. RNA was extracted with RNeasy spin columns (Qiagen, Hilden, Germany) and amplified using the Ovation RNA amplification kit V2 (Nugen Technologies, AC Leek, The Netherlands). RNA was labeled, hybridized and analyzed using NimbleGen (Roche NimbleGen, Madison, WI, USA) Mouse Gene Expression 12 × 135K Arrays according to the manufacturer's recommendations. Each RNA sample (six samples per group for baseline resting condition, and eight samples per group for acute stress condition) was hybridized to two independent NimbleGen slides for technical replication of each measurement.

Quality control and statistical analyses of microarray data

Principal component analysis demonstrated high reproducibility between technical replicates in both experiments and identified one low-quality replicate in baseline resting conditions that was removed from further analyses (the second technical replicate was not affected). Raw data were processed with robust multichip average[25] according to NimbleGen's recommendation. For analysis of differential expression, the expression matrix was log2 transformed and imported into Partek Genomics Studio (Partek, Missouri, USA) and Limma.[26] A linear model was run to identify probes for genes differentially expressed between groups and multiple testing-corrected *P*-values (false discovery rate method) were calculated.

Gene set enrichment analyses

Gene set enrichment analyses (GSEA) algorithm implemented in the GSEA tool from the Broad Institute[27] was used to detect coordinated changes in gene expression in biological networks. This gene set-based approach is particularly powerful for data sets where overall expression analysis proves insensitive.[28] *t*-statistic values (on a gene level) from the analysis of variance model described above were used as a measure of change for all genes whose mouse gene ID could be mapped to a human ortholog, based on the Roche genome annotation infrastructure. The Pathway Commons[29] collection was used as a gene set library, which was limited to pathways containing 5–500 genes. Significantly regulated pathways were identified within GSEA with random permutations in the gene space, the number of permutations was set to 10 000. The lists of significantly up- or downregulated pathways were sorted by false discovery rate-corrected *P*-values, set to the recommended 25% threshold.[27]

Reverse transcription quantitative real-time PCR

Reverse transcription quantitative real-time PCR (RT-qPCR) was performed using SYBR green (Roche Diagnostics, Mannheim, Germany) on a LightCycler II 480 (Roche) according to the manufacturer's recommendations and normalized against Tubulin delta 1. Cycling conditions were 5 min at 95 °C, then 45 cycles with denaturation (10 s at 95 °C), annealing (10 s at 60 °C) and elongation (10 s at 72 °C). Primers were designed using Primer3Plus[30] or Quantprime[31] (see Supplementary Table 2) and tested for quality and specificity by melt-curve analysis, gel electrophoresis and appropriate negative controls.

Bisulfite pyrosequencing

Genomic DNA was extracted and purified using DNeasy Blood and Tissue Kit (Qiagen), and bisulfite treated (EZ DNA Methylation Gold Kit, Zymo Research, Irvine, CA, USA) according to the manufacturer's recommendations. PCR and pyrosequencing primers were designed using Pyromark Assay Design 2.0 (Qiagen). Amplicons containing the *Prkcc* promoter region were generated using a standard PCR protocol, an unmodified forward primer 5'-(AAGATGATTGATTGATTGGGAGAA)-3', and a biotin-labeled reverse primer 5'-(ACACCTAACCATACACAACACAC)-3'. Subsequent pyrosequencing on the PCR amplicon was performed using a PyroMark Q24 Advanced pyrosequencer and appropriate reagents (Pyromark Advanced CpG Reagents, Qiagen) following the manufacturer's recommendations. For high-resolution sequencing, the following two sequencing primers were used (sequencing primer1: 5'-AAGGGGGTGGATAAG-3'; sequencing primer2: 5'-GGGGGTTTTAAATTGAAAT-3'). Average methylation level of CpG sites was quantified using PyroMark Q24 2.0 software (Qiagen).

Electrophysiology

Adult male and female mice were used for electrophysiological experiments. Acute hippocampal slices were prepared and electrophysiological recordings conducted as described previously[32] (for detailed methods see Supplementary Data). For amygdala recordings, horizontal slices were prepared, the recording electrode was placed in the lateral nucleus of the amygdala, one stimulation electrode was placed close to the internal capsule (thalamic pathway), another stimulation electrode was placed externally to the capsule to stimulate fibers from the auditory cortex (cortical pathway).[33] Experimenters were blind to treatment for all experiments. One or more slices from each mouse were used and data were averaged, so that animals and not slices were considered biological replicates.

Behavioral testing

Behavior testing was conducted in adult (3–6-month old) mice under dim red light, and animals were monitored by an experimenter blind to treatment through videotracking from an adjacent room. Forced swim test was conducted in females because our previous data showed stronger depressive-like behavior in F2 females than males. Memory tasks were conducted on males to avoid confounding effects of estrous cycle.[34] See Supplementary Figure S2C for experimental outline.

Forced swim test. Mice were placed in a small tank of water (18 cm high, 13 cm diameter, 18 ± 1 °C, filled up to 12 cm) for 6 min. Floating duration was scored manually.

Novel object recognition. Mice were habituated to an arena (gray plastic box, 25 × 25 × 20 cm) 10 min daily for 3 consecutive days. They were then exposed to three identical unfamiliar objects in the arena for 10 min. After 2.5 h, they were placed back in the arena but one of the objects was replaced with a novel object to test object memory. The following day, the same animals were trained again with different objects and tested 24 h later. The arena floor was lit with infrared light and the animals were tracked with an infrared camera and tracking system (Viewpoint, Lyon, France). Time spent exploring each object was recorded. Animals not exploring each object for at least 5 s during training were excluded from the analyses (two mice excluded in this study).

Contextual fear conditioning. Mice were exposed to a novel context for 3 min in an automated fear conditioning system (TSE, Bad Homburg, Germany), then received two 1-s 0.6-mA foot-shocks 1 min apart. Movements were detected by infrared beams in the testing chamber (TSE). Freezing, defined as the absence of any detectable movement for >1 s, was measured for 4 min in the same context immediately before and 24 h after fear conditioning and served as an indicator of fear memory.

Statistical analyses

For RT-qPCR and pyrosequencing analyses, independent samples *t*-tests were used to compare groups. For electrophysiology, independent samples *t*-tests were used to compare the field excitatory postsynaptic potential slope over the last 20 min of LTP/LTD recordings between groups. For object recognition, analysis of variance was used with object as within-subjects factor. Significant main effects were further analyzed using paired-samples *t*-tests to compare the average time spent exploring the familiar objects and time spent exploring the novel object. Fear conditioning data were analyzed using independent samples *t*-tests. Values were considered outliers if they deviated >2 SDs from the group mean, and this outlier exclusion criterion was pre-established.

RESULTS

To test the effects of early-life stress across generations, we subjected newborn mouse pups (F1) to MSUS for 2 weeks, and then bred the males when adult to naïve (non-stressed) wild-type females to generate F2 progeny. To validate the efficacy of the MSUS manipulation, we examined depressive-like behaviors on a forced swim test. Adult F1 MSUS males and the F2 female offspring spent more time floating than controls (F1: controls: 48.3 ± 6.2; MSUS: 71.0 ± 6.6 s, $t(60) = 2.5$, $P = 0.015$. F2: controls: 46.1 ± 5.6; MSUS: 77.2 ± 7.6 s; $t(28) = 3.30$, $P = 0.003$, Supplementary Figure S3), confirming that MSUS induces depressive-like symptoms across generations.[7–9]

Early life stress is known to lead to cognitive deficits in adulthood,[20,21] therefore we examined memory performance in MSUS mice across generations. We tested contextual fear memory using a paradigm in which a novel context is associated with an aversive stimulus (mild foot-shock). While baseline freezing was similar in control and MSUS mice from F1, F2 or F3 generations (Figure 1a–c, left), F1 MSUS mice spent significantly less time freezing than controls 24 h after fear conditioning (controls: $55.6 \pm 2.1\%$; MSUS: $44.8 \pm 3.7\%$; $t(18) = 2.52$, $P = 0.021$, Figure 1a). F2 MSUS offspring had similar lower freezing 24 h after conditioning (controls: $70.7 \pm 2.3\%$; MSUS: $59.6 \pm 3.4\%$; $t(22) = 2.66$, $P = 0.014$, Figure 1b). In contrast, freezing was normal in F3 MSUS offspring (controls: $56.4 \pm 3.2\%$; MSUS: $60.2 \pm 3.1\%$; $t(28) = 0.82$, $P = 0.417$; Figure 1c), suggesting that MSUS impairs contextual fear memory in animals directly exposed to MSUS (F1) and their offspring (F2), but not in the following generation (F3). To examine if MSUS also impairs performance in another memory test that recruits the hippocampus, we assessed memory on a novel object recognition task. The animals were trained to memorize several objects in a familiar arena, then their memory for the objects was evaluated 2.5 or 24 h later.[35] During training, F1 control and MSUS mice explored all objects equally, suggesting similar acquisition (data not shown). When tested 2.5 h later, both groups spent more time exploring a novel object than the familiar objects (Controls: $t(7) = 4.11$, $P = 0.004$; MSUS: $t(12) = 3.71$, $P = 0.003$; Figure 2a), indicating normal object memory. However, 24 h after training, while control mice spent more time exploring the novel object ($t(9) = 4.43$, $P = 0.002$), F1 MSUS mice could not discriminate it from the familiar objects ($t(10) = 1.60$, $P = 0.14$, Figure 2b), indicating impaired long-term object memory. Similarly to F1 animals, F2 MSUS mice had normal object memory 2.5 h after training (Controls: $t(15) = 3.90$, $P = 0.003$; MSUS: $t(13) = 3.46$, $P = 0.004$; Figure 2c), but impaired memory after 24 h (Controls: $t(15) = 3.90$, $P = 0.001$); MSUS: $t(14) = 1.65$, $P = 0.12$, Figure 2d). Together, these data suggest that MSUS impairs memory across generations.

To determine which molecular pathways are affected by MSUS, we conducted genome-wide DNA microarray analyses in adult F2 females at rest (baseline resting condition). In the hippocampus, a brain area implicated in depression and memory,[18,36] 156 genes were differentially regulated by at least 1.2-fold in F2 MSUS mice (data available through GEO, accession number GSE47848). However, because multiple testing correction (false discovery rate method) proved too stringent for the data set (see Methods), we used a gene set-based approach to determine whether gene expression was altered at the network level.[28] GSEA identified 49 upregulated and 30 downregulated molecular pathways in F2 MSUS hippocampus compared with controls (see Supplementary Table 1 for a complete list). Notably, the downregulated pathways contained several partially overlapping components critical for excitatory synaptic transmission such as *N*-Methyl-D-aspartate (NMDA) receptor-dependent signaling, neuronal plasticity and memory formation (Table 1, top).

MSUS was previously shown to alter behavioral responses in stressful and aversive conditions in adult animals across generations.[8,10] We postulated that differences in gene expression between MSUS and control mice could be more pronounced after a stress challenge. We repeated the DNA microarray analyses in F2 animals 45 min after exposure to a session of forced swim, an acute form of stress that activates gene expression in the hippocampus.[37] Using the same statistical criteria as above, 1782 genes with differential expression could be identified in F2 MSUS mice after acute stress, representing a 10-fold increase compared with baseline resting conditions. GSEA identified 25 gene pathways significantly downregulated in MSUS mice, but no upregulated pathway (Supplementary Table 1). Notably, the downregulated pathways included NMDA receptor-dependent signaling, synaptic calcium signaling and synaptic plasticity networks, which are similar to those identified in resting conditions (Table 1). Overall, these results show that plasticity-related gene networks are compromised in the offspring of MSUS mice, both at rest and after and acute stress challenge.

An alteration of signaling pathways can dramatically affect neuronal and network functions. The broad and coordinated molecular changes detected in plasticity-related pathways in F2 MSUS mice may therefore have important functional consequences. We tested this possibility by examining synaptic plasticity in different brain regions in adult F2 mice. In hippocampus area CA1, LTP induced by one train of 100 Hz stimulation, a form of synaptic strengthening, was abolished in MSUS slices and instead, synaptic depression was induced. This

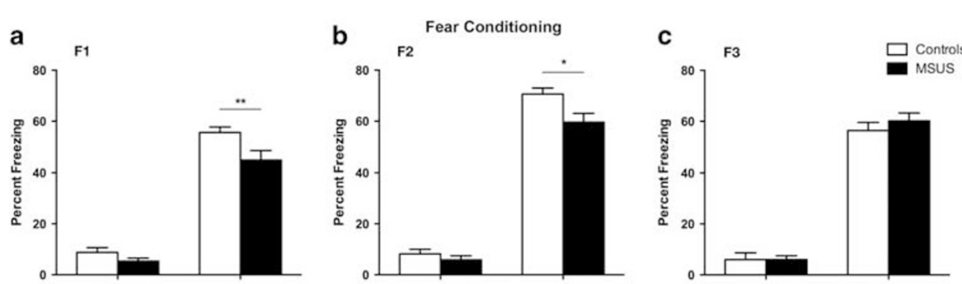


Figure 1. Impaired fear memory in unpredicatable maternal seperation cambined with unpredictable material strees (MSUS) mice across generations. Twenty-four hours after contextual fear conditioning, F1 (**a**) and F2 (**b**) MSUS mice spend less time freezing than controls (F1: controls, $n = 10$; MSUS, $n = 13$. F2: controls, $n = 12$; MSUS, $n = 12$). No differences in freezing are detected between F3 MSUS and control mice (**c**) (controls, $n = 12$; MSUS, $n = 18$). Left bars show baseline freezing before delivery of the foot shock, right bars show freezing during context test. Data are mean $\pm$ s.e.m. $^{**}P < 0.01$, $^{*}P < 0.05$.

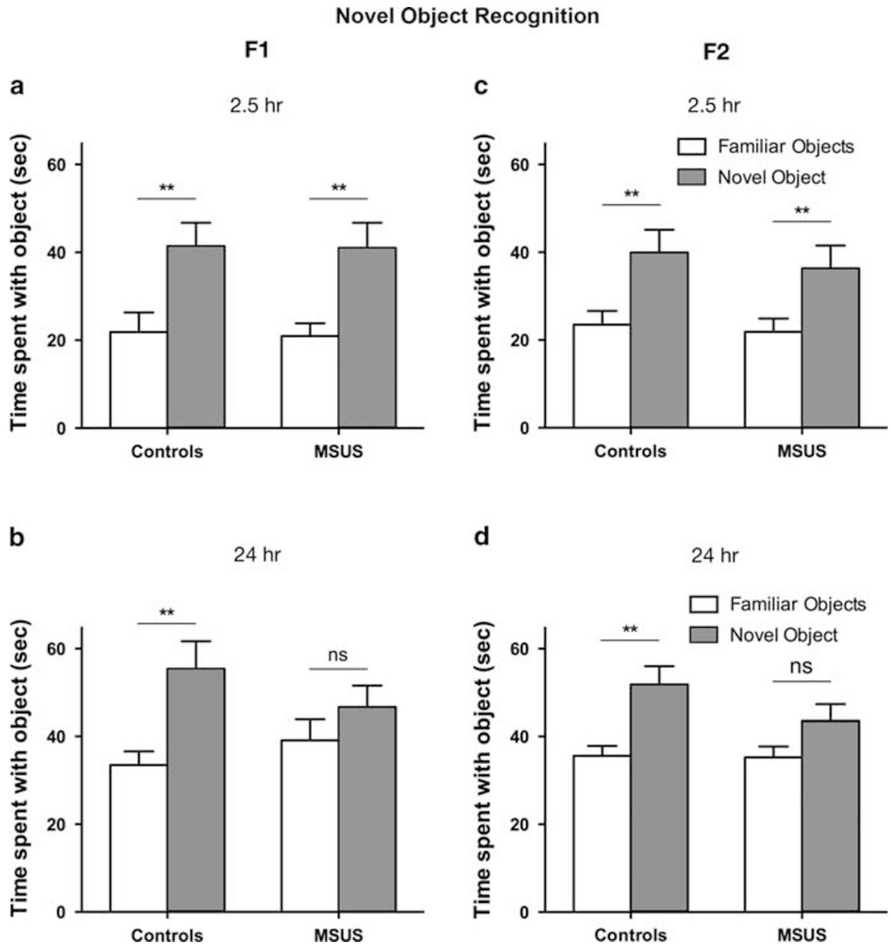


Figure 2. Impaired object recognition memory in unpredictable maternal separation combined with unpredictable maternal stress (MSUS) mice across generations. When tested 2.5 h after training, control mice and MSUS mice spend more time exploring the novel object than the familiar objects, both (**a**) in F1 animals (controls: $n = 10$; MSUS: $n = 13$), and (**c**) the F2 offspring (controls: $n = 15$; MSUS: $n = 15$). Twenty-four hours after training, control mice spend more time exploring the novel object, but MSUS mice do not discriminate between novel and familiar objects, both (**b**) in F1 animals (controls: $n = 10$; MSUS: $n = 11$) and (**d**) F2 offspring (controls: $n = 16$; MSUS: $n = 15$). Data are mean ± s.e.m. $^{**}P < 0.01$, $^{*}P < 0.05$.

effect persisted throughout the recording and was the most pronounced 40–60 min after the tetanus in both, males and females (males: controls, 135.3 ± 12.1%; MSUS: 43.05 ± 4.6%; $t(7) = 7.86$, $P < 0.001$. Females: controls, 126.0 ± 13.6%; MSUS: 65.81 ± 14.4%; $t(8) = 3.04$, $P = 0.016$, Figures 3a and b). Two-pathway recordings confirmed that the observed depression in F2 MSUS mice was not due to a reduced viability of slices, since a stable response could be induced by basal stimulation in a non-tetanized pathway (Supplementary Figure S4). We next examined whether a stronger stimulation could elicit LTP in MSUS slices and used three trains of 100 Hz tetanus, a stimulation known to produce late-phase LTP.[38] Like one-train LTP, three-train LTP was abolished in F2 MSUS slices compared with control slices (40–60 min post tetanus, controls: 153.3 ± 13.4%; MSUS: 98.0 ± 12.7%; $t(10) = 2.98$, $P = 0.013$; Figure 3c), even when the stimulation was repeated twice (LTP after 1st tetanus: controls 183 ± 15.8%; MSUS: 128.3 ± 7.5%, $t(8) = 2.66$; $P = 0.029$. LTP after 2nd tetanus: controls 290.9 ± 40.79%; MSUS: 134.3 ± 13.4%; $t(8) = 3.01$; $P = 0.017$, Figure 3d). We next examined LTD, a form of synaptic weakening induced by low-frequency stimulation. In hippocampus area CA1, LTD was stronger in F2 MSUS slices than in control slices (controls: 74.9 ± 2.9%; MSUS: 63.0 ± 3.7%; $t(6) = 2.50$, $P = 0.046$; Figure 3e). The changes in plasticity in F2 MSUS hippocampus were not due to any gross alteration in basal synaptic transmission since input–output curves were similar in control and MSUS slices (Figure 3f). They were also not due to any major alteration in neurotransmitter release since paired pulse facilitation, a short-term form of presynaptic plasticity,[39] was comparable in control and MSUS slices (Figure 3g). We then tested if plasticity was disrupted in other brain areas and examined LTP in the lateral amygdala, a part of the limbic system implicated in fear memory. Stimulation of the thalamic or cortical pathways in lateral amygdala[33] showed that LTP was impaired in both pathways in MSUS slices (thalamic pathway: controls: 181.1 ± 15.3%; MSUS: 115.1 ± 19.4%; $t(11) = 2.61$, $P = 0.024$; cortical pathway: controls: 205.6 ± 52.9%; MSUS: 110.8 ± 12.7%; $t(10) = 2.05$, $P = 0.068$, Supplementary Figure S5). Together, these results indicate a global alteration of synaptic plasticity in several brain areas in the F2 offspring, with a shift in plasticity toward synaptic depression in the adult hippocampus.

To determine if the LTP impairment in the offspring was inherited from the fathers, we examined LTP in the hippocampus of F1 MSUS males. Three-train LTP was abolished in F1 MSUS males (controls: 193.5 ± 16.7%; MSUS: 112.7 ± 13.1%; $t(9) = 3.87$, $P = 0.004$; Figure 4a), similarly to that in F2 offspring, suggesting that the LTP defect was transmitted from father to offspring. We then tested if transmission depends on maternal care by conducting cross-fostering. When adult, F2 MSUS pups raised

Table 1. Plasticity-related pathways identified as downregulated in the hippocampus of F2 MSUS mice relative to controls by gene set enrichment analysis.

Rank	*Gene set name*	*Size*	*NES*	*Nom. P-value*	*FDR q-value*
Brain plasticity pathways downregulated in F2 MSUS mice—baseline resting condition					
5	Reelin signaling pathway—NCI/nature pathway	28	− 1.799	0.0009	0.1130
6	Unblocking of NMDA receptor glutamate binding and activation—Reactome pathway	15	− 1.745	0.0026	0.2058
8	Synaptic transmission—Reactome pathway	150	− 1.585	0.0012	0.2172
14	CREB phosphorylation through the activation of CaMKII—Reactome pathway	13	− 1.697	0.0055	0.2318
17	CREB phosphorylation through the activation of RAS—Reactome pathway	23	− 1.547	0.0228	0.2394
19	Activation of NMDA receptor upon glutamate binding and postsynaptic events—Reactome pathway	33	− 1.567	0.0137	0.2395
20	RAS activation upon Ca^{2+} influx through NMDA receptor—Reactome pathway	15	− 1.673	0.0075	0.2400
30	Glutamate binding—Activation of AMPA receptors and synaptic plasticity—Reactome pathway	28	− 1.677	0.0058	0.2479
Brain plasticity pathways downregulated in F2 MSUS mice—acute stress condition					
1	DARPP 32 events—Reactome pathway	23	− 2.356	0.0000	0.0082
3	RAS activation upon Ca^{2+} influx through NMDA receptor—Reactome pathway	14	− 2.169	0.0006	0.0278
4	GABA synthesis, release, reuptake and degradation—Reactome pathway	19	− 2.101	0.0000	0.0459
8	CREB phosphorylation through the activation of CaMKII—Reactome pathway	13	− 1.930	0.0036	0.1428
13	AKT phosphorylates targets in the cytosol—Reactome pathway	13	− 1.867	0.0073	0.1556
20	PI3K-AKT activation—Reactome pathway	36	− 1.779	0.0024	0.2050
21	Transmission across chemical synapses—Reactome pathway	173	− 1.769	0.0000	0.2095

Abbreviations: CREB, cAMP response element-binding protein; DARPP, dopamine- and cAMP-regulated neuronal phosphoprotein; FDR, false discovery rate; GABA, gamma-amino butyric acid; MSUS, unpredictable maternal separation combined with unpredictable maternal stress; NES, normalized enrichment score; NMDA, *N*-Methyl-D-aspartate. Pathways involved in synaptic plasticity are ranked by false discovery rate (FDR) adjusted *P*-value. Rank = rank of the pathway in the complete set of downregulated pathways. Size = number of genes in each pathway. Nom. *P*-value = nominal (unadjusted) *P*-value.

by control dams (F2 MSUS-CD) had impaired three-train LTP, while F2 control pups raised by dams mated to MSUS males (F2 controls-MD) had normal LTP (MSUS-CD: 117.6 ± 7.2%; $t(16) = 4.31$, control-MD: 181.8 ± 12.0%; $P < 0.001$, Figure 4b). These results demonstrate that the negative effects of paternal stress on LTP are transmitted to the offspring via a route independent of maternal care, likely involving the male germline. Finally, we tested whether the LTP defect could also be transmitted to the F3 offspring and for this, we bred F2 males to naïve females. Three-train LTP was robust in both controls and MSUS F3 mice (controls: 214.7 ± 13.6%; MSUS: 177.3 ± 18.1%; $t(11) = 1.604$, $P = 0.137$, Figure 4c), indicating that MSUS affects plasticity in directly exposed mice (F1) and their progeny (F2), but not in the following generation (F3).

We then further examined some of molecular targets relevant for synaptic plasticity found to be altered by DNA microarrays. Using RT-qPCR, we observed that several genes critical for synaptic plasticity including protein kinase C gamma (*Prkcc*; $t(13) = 3.03$, $P = 0.009$), NR1 subunit of the NMDA receptor (*Grin1*; $t(13) = 2.46$; $P = 0.028$), metabotropic glutamate receptor 1 (*Grm1*; $t(13) = 2.26$, $P = 0.044$), calcium/calmodulin-dependent protein kinase II alpha (*Camk2a*; $t(13) = 1.88$, $P = 0.082$) and ionotropic glutamate receptor AMPA3 (*Gria3*; $t(13) = 1.85$, $P = 0.088$) were downregulated in the hippocampus of F2 MSUS mice (Figure 5a, left panel). Several housekeeping genes including *Hprt, Actb* and *Gapdh* were not altered (Figure 5a, right panel), suggesting pathway-specific gene alteration. Because *Prkcc* is a neuron-specific isoform of PKC involved in LTP induction and memory processes,[22,23] we focused on this target and investigated the potential mechanisms underlying its altered expression. We examined whether DNA methylation, an epigenetic mode of gene regulation implicated in the expression and transmission of the effects of MSUS,[8] is affected at the Prkcc promoter. Using bisulfite pyrosequencing, we quantified DNA methylation in a proximal promoter region of *Prkcc* that carries transcription factor-binding sites sufficient for promoter activity[40,41] (Figure 5b). DNA methylation was low across CpGs contained in this region, except at CpG 6, an Sp1-binding site that can bind the transcriptional repressor and/or activator Sp1 and Sp3 with equal affinity.[42] At this site, DNA methylation was significantly lower in F2 MSUS hippocampus compared to controls ($t(10) = 3.30$; $P = 0.008$; Figure 5c). Likewise, in sperm samples of F1 mice, DNA methylation was low at most CpGs except at CpG 6, and like in the hippocampus, it was significantly downregulated at this site in MSUS samples ($t(12) = 2.34$, $P = 0.038$, Figure 5d). In agreement with the observation that no memory or plasticity impairments are transmitted to the F3 generation, we did not observe any difference in DNA methylation at the *Prkcc* promoter in sperm cells of F2 males, nor in the hippocampus of their F3 offspring (Supplementary Figure S6). Interestingly, however, in F1 hippocampus, where LTP and memory are impaired, DNA methylation of the *Prkcc* promoter was not altered and *Prkcc* expression was normal (Supplementary Figure S7). This suggests that molecular changes induced by MSUS in F1 sperm and F2 brain are specific.

DISCUSSION

Using a model of chronic and unpredictable traumatic stress in early life in mice, we demonstrate for the first time that such stress dramatically alters synaptic plasticity in different brain areas and impairs long-term memory in both, the animals directly exposed to stress and their offspring. These defects are associated with changes in several molecular pathways involved in plasticity and memory, and with specific components of these pathways such as *Prkcc*. We show that *Prkcc* expression is altered by traumatic stress in the hippocampus of the offspring and that DNA methylation in *Prkcc* promoter is reduced in both the hippocampus of the offspring, and the sperm of fathers.

Early life stress is known to impair neuronal plasticity and cognitive functions during adulthood in rodents and humans[20,43,44] in part through perturbations in glutamatergic pathways downstream of NMDA receptors.[45–47] However, transgenerational effects on synaptic plasticity have not been examined so far. Our findings that several glutamatergic signaling networks are altered in the hippocampus of the progeny of MSUS males demonstrate that the impact of stress on plasticity pathways is specific and transgenerational. Two of the molecular pathways suppressed by MSUS, both at rest and following acute stress, are 'calcium-mediated RAS activation through NMDA receptors', and

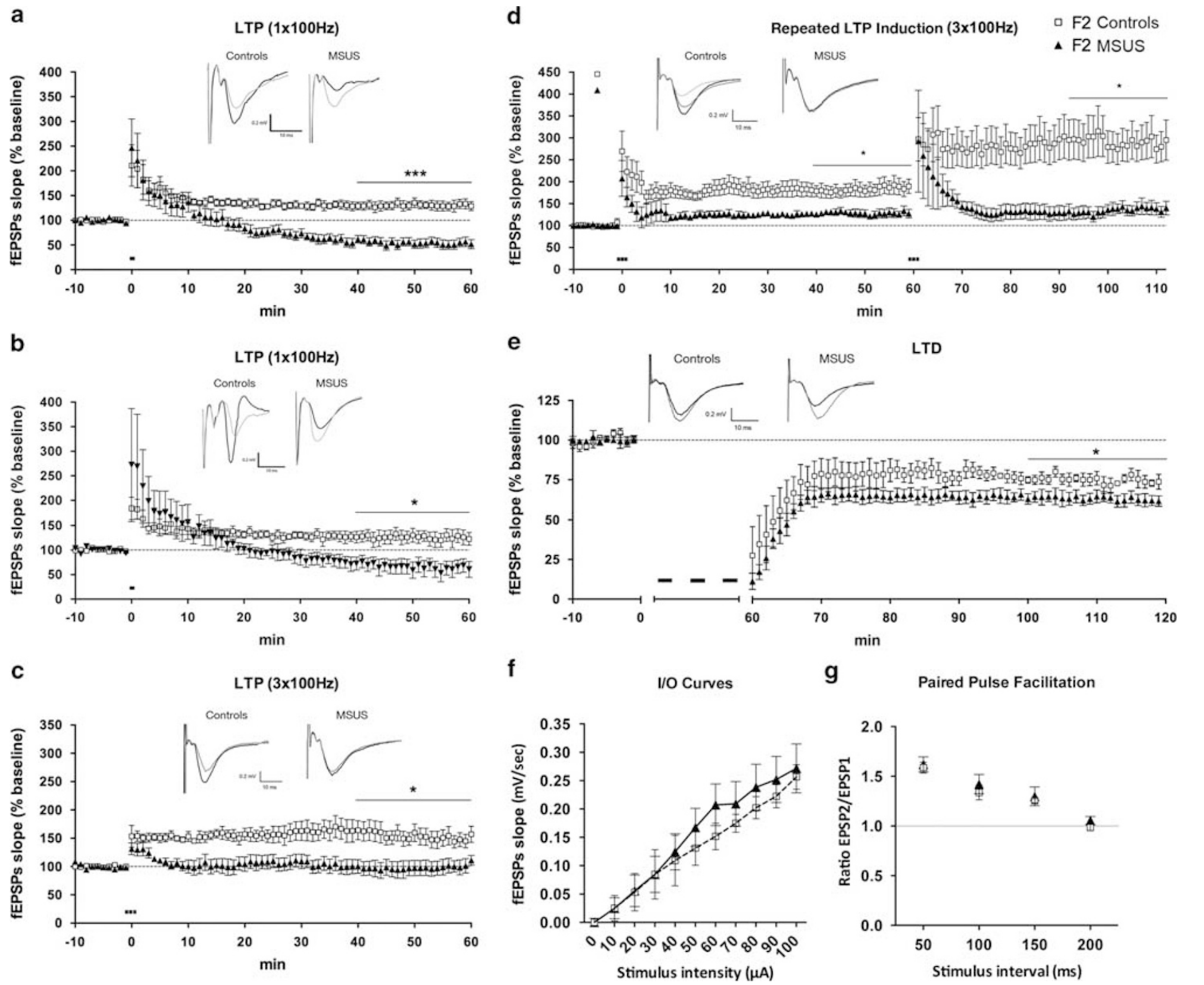


Figure 3. The offspring of unpredictable maternal separation combined with unpredicatable maternal stress (MSUS) males have altered hippocampal long-term potentiation (LTP) and long-term depression (LTD). (**a**) Hippocampal LTP induced by 1 × 100-Hz stimulation is impaired in F2 MSUS males ($n = 5$ mice) compared with controls ($n = 4$ mice). (**b**) A similar impairment is observed in F2 MSUS females ($n = 5$ mice) compared with controls ($n = 5$ mice). (**c**) LTP induced by 3 × 100-Hz stimulation leads to impaired LTP in F2 MSUS mice ($n = 6$ mice) compared with controls ($n = 6$ mice). (**d**) After repeated 3 × 100-Hz stimulation, the LTP impairment persists in MSUS mice ($n = 4$ mice) compared with controls ($n = 6$ mice). (**e**) LTD is enhanced in F2 MSUS mice ($n = 4$ mice) compared with controls ($n = 4$ mice). (**f**) Input–output curves are comparable in F2 MSUS mice ($n = 6$ mice) and controls ($n = 6$ mice). (**g**) Paired pulse facilitation is comparable in F2 MSUS mice ($n = 5$ mice) and controls ($n = 6$ mice). Thick black bars schematically indicate the LTP/LTD stimulation protocol. Horizontal bars indicate the last 20 min of recording used for statistical analyses. Inset traces show the field excitatory postsynaptic potential (fEPSP) from a representative animal from each group, gray traces represent the average of baseline recording before stimulation, black traces show the average over the last 20 min of post-stimulation recording. Data are mean ± s.e.m. $^{***}P < 0.001$, $^{**}P < 0.01$, $^{*}P < 0.05$.

'CaMKII-dependent cAMP response element-binding protein phosphorylation' pathways. These pathways are critical for the induction and the maintenance of LTP,[48,49] and may therefore underlie the shift in hippocampal plasticity of the offspring. Such a shift was previously reported in response to acute stress[50–53] or chronic social defeat stress,[54] but our data newly show that it occurs also in the progeny of animals subjected to stress. It is reminiscent of the synaptic modification model of homeostasis that proposes that neuronal networks can adapt to repeated strengthening or weakening of synapses.[54–57] In turn, it suggests that early-life stress may modulate the synaptic range in a way that further strengthening of the synapse becomes difficult, while synaptic weakening is facilitated. Such a shift in synaptic range, as recently observed in adult mice previously exposed to maternal separation,[58] may explain why the MSUS offspring have a relatively mild memory deficit despite a total absence of LTP. The altered expression of plasticity-related genes in MSUS offspring including *Prkcc*, likely underlies these defects. Indeed, knockout mice deficient for *Prkcc* have a phenotype remarkably similar to MSUS mice; they lack LTP[23] and have mild memory impairments.[22] However, *Prkcc* is only one of several affected targets in MSUS animals, and the phenotype likely results from the combined action of all altered genes. Network-wide expression changes were recently reported in the paraventricular nucleus and the bed nucleus of stria terminalis in the offspring of stressed fathers.[59] Together, these data suggest that broad transcriptomic changes occur in the offspring of stressed fathers in several brain regions and may subserve different functions. Importantly, the

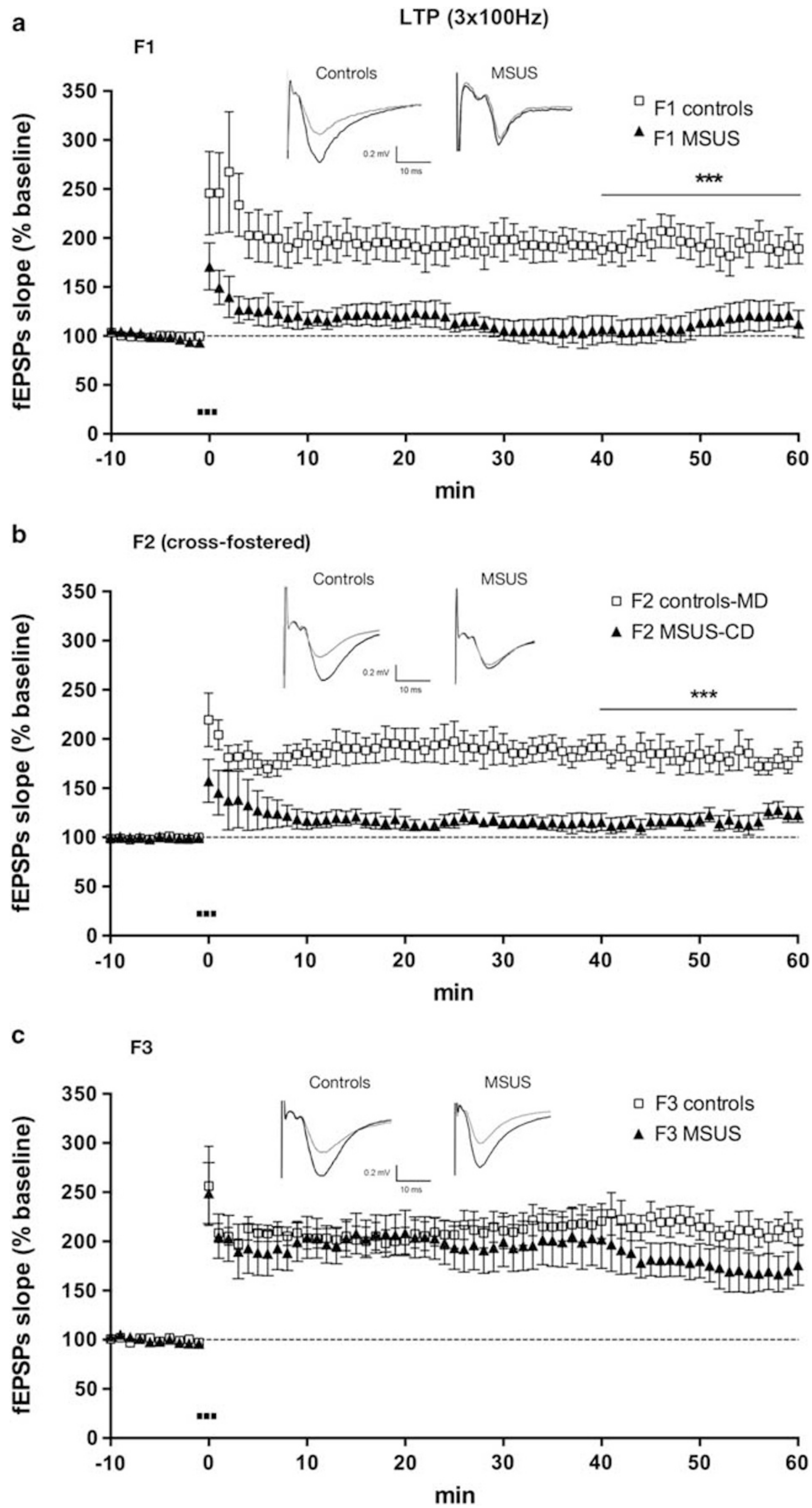
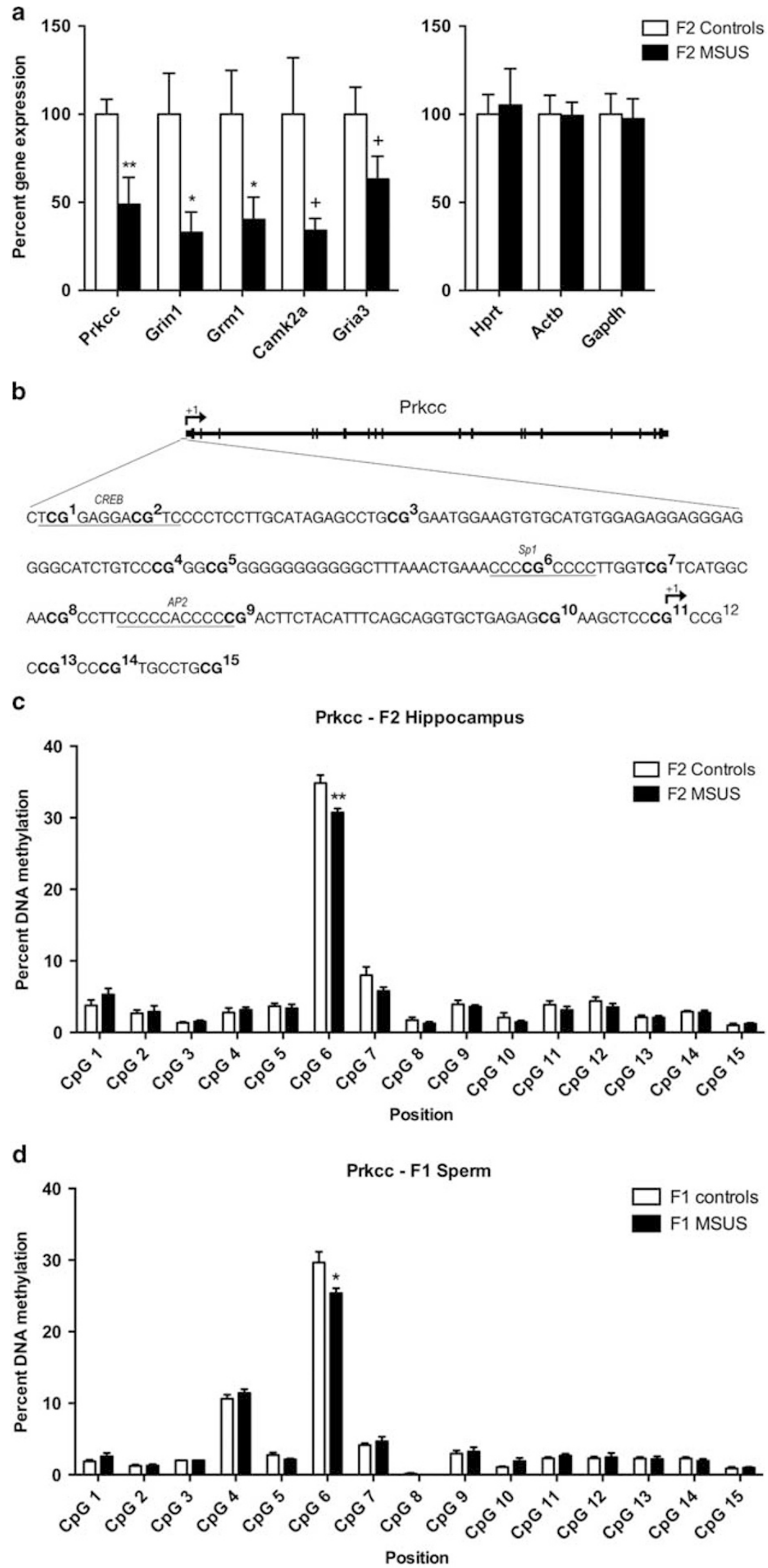


Figure 4. Impaired long-term potentiation (LTP) in unpredictable maternal separation combined with unpredictable maternal stress (MSUS) mice across generations. (**a**) LTP impairment is observed in F1 MSUS mice (n=6 mice) compared to controls ($n=6$ mice). (**b**) A similar LTP impairment is observed in MSUS offspring (F2 generation) raised by control dams after cross-fostering (MSUS-CD; $n=8$ mice, 3 litters per group were used for cross-fostering). Control offspring raised by dams previously mated with MSUS males (control-MD; $n=10$ mice) have intact hippocampal LTP. (**c**) In the F3 generation, LTP can be similarly induced in MSUS mice ($n=7$ mice) and controls ($n=6$ mice). Horizontal bars indicate the last 20 min of recording used for statistical analyses. Inset traces show a typical field excitatory postsynaptic potential (fEPSP) from each group, gray traces represent the average of baseline recording before stimulation, black traces show the average over the last 20 min of post-stimulation recording. Data are mean $\pm$ s.e.m. ***$P < 0.001$.



finding that *Prkcc* expression and methylation is not altered in the hippocampus of F1 MSUS mice suggests that the molecular mechanisms underlying the LTP and memory impairments in F1 and F2 mice may differ considerably in both generations. Maternal separation paradigms affect many proteins critical for synaptic plasticity in the animals directly exposed to stress, including NMDA and AMPA receptor subunits,[46] NCAM[60] or adenosine A2a receptors,[61] targets that noticeably were not altered in our

Figure 5. Unpredictable maternal separation combined with maternal stress (MSUS) alters *Prkcc* expression and DNA methylation in the *Prkcc* promoter in F1 sperm and F2 brain. (**a**) Left: reverse transcription quantitative real-time PCR (RT-qPCR) confirms decreased expression of genes related to synaptic plasticity and memory in the hippocampus of F2 MSUS mice ($n = 8$) compared with controls ($n = 8$). Right: no group difference is detected for housekeeping genes. (**b**) Schematic representation of the *Prkcc* promoter region analyzed by pyrosequencing including the transcription start site (+1) and the binding site for cAMP response element-binding protein (CREB), Sp1 and AP2. Numbers represent individual CpG sites analyzed for DNA methylation (CpG 1–15). (**c**) DNA methylation is reduced at CpG 6 in the hippocampus of F2 MSUS mice compared with controls ($n = 6$/group). (**d**) DNA methylation is reduced at CpG 6 in the sperm of F1 MSUS mice ($n = 7$) compared with controls ($n = 8$).

microarray analyses of MSUS offspring. Therefore, early-life stress seems to induce different molecular changes in the brain and in germ cells of the stressed mice (F1), yet these different routes seem to give rise to similar plasticity and memory impairments in F1 and F2 MSUS mice. Further, a recent study found that chronic social instability during adolescence can lead to a transgenerational increase in anxiety and social deficits, linked to yet another calcium-activated signaling molecule, *Rcan1/Rcan2*.[62] Therefore, different forms of chronic stress seem to be able to target different synaptic plasticity pathways in the offspring in a highly specific manner.

We have previously described transgenerational impairments in social recognition memory following MSUS.[9] The current results extend these findings by showing that cognitive functions are also affected. Interestingly, while our data show the negative impact of early traumatic stress on both plasticity and cognitive functions across generations, another study showed that stimulating environmental conditions can positively modulate plasticity and cognition.[63] If mice were transiently exposed to enriched environmental conditions early in life, their offspring had enhanced LTP and better contextual fear memory. These effects persisted through cross-fostering. Although the effect of enrichment was transmitted by females and not males, this suggests that plasticity and memory are sensitive to parental experience and can be impaired or enhanced in the offspring, depending on the environment encountered by the parents. In this context, our study shows that both females and males have altered plasticity, while previous studies have reported sex-specific effects in the offspring.[8,64] These results suggest that MSUS has a global impact in the offspring independent of sexual programming.

Paternal effects of environmental factors including stress,[8,65,66] endocrine disruptors,[67,68] diet[69,70] and drugs of abuse[71] on the offspring have been documented but the mechanisms underlying transmission remain poorly understood. In our model, since the transmitting males contribute only their germ cells and are never in contact with the offspring,[12] transmission most likely implicates epigenetic mechanisms in the germline. The altered DNA methylation in the sperm of MSUS fathers, that is, at the *Prkcc* promoter, provides evidence that DNA methylation in germ cells is associated with transmission. This complements our previous demonstration of DNA hypo- or hyper-methylation at several loci in MSUS F1 sperm and in the brain of F2 offspring.[8] Such alterations may be maintained or relayed by other mechanisms in the developing embryo (which undergoes widespread demethylation) and contribute to the adult phenotypes.[72] How DNA methylation is modified by early-life experiences in sperm cells, and how changes are targeted to specific loci remain unknown. Although male germ cells are the primary carrier, maternal care may also contribute to the transmission of paternal effects, since females can adjust their level of care depending on the fitness and attractiveness of their mate.[73,74] This possibility is, however, excluded for the MSUS model since the impairments (LTP) persist after cross-fostering. Interestingly, while emotional reactivity, depressive-like behaviors and social behaviors due to MSUS are transmitted down to the F3 generation,[8,9] the LTP and memory impairments are transmitted only to F2 and do not affect F3 animals. This suggests that different mechanisms may be recruited for transmission, some that persist across multiple generations and some that are more transient.[11,75] Indeed, we have recently also demonstrated a causal role of small non-coding RNAs in transmission of stress-induced phenotypes across generations,[10] emphasizing that more work is needed to dissect how these different mechanisms contribute to transmission of stress-induced traits across generations.

The present findings in mice are expected to have important consequences in humans, since cognitive ability and intelligence, although known to be highly heritable traits,[76] still have no clear genetic basis.[77,78] Such 'missing heritability' of complex traits is classically postulated to be due to gene–gene and gene–-environment interactions involving multiple, often rare gene variants that bring small effects.[79] Our results suggest that environmental factors encountered by parents also contribute to the heritability of cognitive abilities. Such a link is difficult to study in humans due to the complexity of the genome.[75] Animal models like ours therefore provide a valuable means to study the underlying mechanisms, and gain novel insight with potential implication for the clinic.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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REFERENCES

1 Schafer I, Fisher HL Childhood trauma and psychosis—what is the evidence?. *Dialogues Clin Neurosci* 2011; **13**: 360–365.
2 Norman RE, Byambaa M, De R, Butchart A, Scott J, Vos T The long-term health consequences of child physical abuse, emotional abuse, and neglect: a systematic review and meta-analysis. *PLoS Med* 2012; **9**: e1001349.
3 U.S. Department of Health and Human Services Administration on Children, Youth and Families, Children's Bureau. (2012). Child Maltreatment 2011.
4 Gilbert R, Widom CS, Browne K, Fergusson D, Webb E, Janson S Burden and consequences of child maltreatment in high-income countries. *Lancet* 2009; **373**: 68–81.
5 Lampe A The prevalence of childhood sexual abuse, physical abuse and emotional neglect in Europe. *Z Psychosom Med Psychother* 2002; **48**: 370–380.
6 Schönbucher V, Maier T, Held L, Mohler-Kuo M, Schnyder U, Landolt MA Prevalence of child sexual abuse in Switzerland: a systematic review. *Swiss Med Wkly* 2011; **140**: w13123.
7 Weiss IC, Franklin TB, Vizi S, Mansuy IM Inheritable effect of unpredictable maternal separation on behavioral responses in mice. *Front Behav Neurosci* 2011; **5**: 3.
8 Franklin TB, Russig H, Weiss IC, Gräff J, Linder N, Michalon A *et al.* Epigenetic transmission of the impact of early stress across generations. *Biol Psychiatry* 2010; **68**: 408–415.
9 Franklin TB, Linder N, Russig H, Thöny B, Mansuy IM Influence of early stress on social abilities and serotonergic functions across generations in mice. *PLoS ONE* 2011; **6**: e21842.

10 Gapp K, Jawaid A, Sarkies P, Bohacek J, Pelczar P, Prados J *et al.* Implication of sperm RNAs in transgenerational inheritance of the effects of early trauma in mice. *Nat Neurosci* 2014;**17**: 667–669.
11 Bohacek J, Mansuy IM Epigenetic inheritance of disease and disease risk. *Neuropsychopharmacology* 2013; **38**: 220–236.
12 Bohacek J, Gapp K, Saab BJ, Mansuy IM Transgenerational epigenetic effects on brain functions. *Biol Psychiatry* 2012; **73**: 313–320.
13 Pechtel P, Pizzagalli DA Effects of early life stress on cognitive and affective function: an integrated review of human literature. *Psychopharmacology* 2011; **214**: 55–70.
14 Arts B, Jabben N, Krabbendam L, van Os J Meta-analyses of cognitive functioning in euthymic bipolar patients and their first-degree relatives. *Psychol Med* 2008; **38**: 771–785.
15 McClintock SM, Husain MM, Greer TL, Cullum CM Association between depression severity and neurocognitive function in major depressive disorder: a review and synthesis. *Neuropsychology* 2010; **24**: 9–34.
16 Gould F, Clarke J, Heim C, Harvey PD, Majer M, Nemeroff CB The effects of child abuse and neglect on cognitive functioning in adulthood. *J Psychiatr Res* 2012; **46**: 500–506.
17 McEwen BS, Gianaros PJ Central role of the brain in stress and adaptation: links to socioeconomic status, health, and disease. *Ann NY Acad Sci* 2010; **1186**: 190–222.
18 Kim JJ, Diamond DM The stressed hippocampus, synaptic plasticity and lost memories. *Nat Rev Neurosci* 2002; **3**: 453–462.
19 Wang X-D, Rammes G, Kraev I, Wolf M, Liebl C, Scharf SH *et al.* Forebrain CRF1 modulates early-life stress-programmed cognitive deficits. *J Neurosci* 2011; **31**: 13625–13634.
20 Ivy AS, Rex CS, Chen Y, Dubé C, Maras PM, Grigoriadis DE *et al.* Hippocampal dysfunction and cognitive impairments provoked by chronic early-life stress involve excessive activation of CRH receptors. *J Neurosci* 2010; **30**: 13005–13015.
21 Murgatroyd C, Patchev AV, Wu Y, Micale V, Bockmühl Y, Fischer D *et al.* Dynamic DNA methylation programs persistent adverse effects of early-life stress. *Nat Neurosci* 2009; **12**: 1559–1566.
22 Abeliovich A, Paylor R, Chen C, Kim JJ, Wehner JM, Tonegawa S.PKC gamma mutant mice exhibit mild deficits in spatial and contextual learning. *Cell* 1993; **75**: 1263–1271.
23 Abeliovich A, Chen C, Goda Y, Silva AJ, Stevens CF, Tonegawa S Modified hippocampal long-term potentiation in PKC gamma-mutant mice. *Cell* 1993; **75**: 1253–1262.
24 Hickman DL, Swan MP Effects of age of pups and removal of existing litter on pup survival during cross-fostering between multiparous outbred mice. *J Am Assoc Lab Anim Sci* 2011; **50**: 641–646.
25 Bolstad BM, Irizarry RA, Astrand M, Speed TP A comparison of normalization methods for high density oligonucleotide array data based on variance and bias. *Bioinformatics* 2003; **19**: 185–193.
26 Smyth GK *Bioinformatics and Computational Biology Solutions using R and Bioconductor* (Gentleman R, Carey V, Dudoit S, Irizarry RA, Huber W). Springer, 2005, pp 397–420.
27 Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA *et al.* Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci USA* 2005; **102**: 15545–15550.
28 Nam D, Kim SY Gene-set approach for expression pattern analysis. *Brief Bioinform* 2008; **9**: 189–197.
29 Cerami EG, Gross BE, Demir E, Rodchenkov I, Babur O, Anwar N *et al.* Pathway Commons, a web resource for biological pathway data. *Nucleic Acids Res* 2011; **39**: D685–D690.
30 Untergasser A, Nijveen H, Rao X, Bisseling T, Geurts R, Leunissen JA Primer3Plus, an enhanced web interface to Primer3. *Nucleic Acids Res* 2007; **35**: W71–W74.
31 Arvidsson S, Kwasniewski M, Riaño-Pachón DM, Mueller-Roeber B QuantPrime—a flexible tool for reliable high-throughput primer design for quantitative PCR. *BMC Bioinformatics* 2008; **9**: 465.
32 Knobloch M, Farinelli M, Konietzko U, Nitsch RM, Mansuy IM Abeta oligomer-mediated long-term potentiation impairment involves protein phosphatase 1-dependent mechanisms. *J Neurosci* 2007; **27**: 7648–7653.
33 Humeau Y, Shaban H, Bissiere S, Luthi A Presynaptic induction of heterosynaptic associative plasticity in the mammalian brain. *Nature* 2003; **426**: 841–845.
34 Daniel JM, Bohacek J The critical period hypothesis of estrogen effects on cognition: insights from basic research. *Biochim Biophys Acta* 2010; **1800**: 1068–1076.
35 Genoux D, Haditsch U, Knobloch M, Michalon A, Storm D, Mansuy IM Protein phosphatase 1 is a molecular constraint on learning and memory. *Nature* 2002; **418**: 970–975.
36 Small SA, Schobel SA, Buxton RB, Witter MP, Barnes CA A pathophysiological framework of hippocampal dysfunction in ageing and disease. *Nat Rev Neurosci* 2011; **12**: 585–601.
37 Gray JD, Rubin TG, Hunter RG, McEwen BS. Hippocampal gene expression changes underlying stress sensitization and recovery. *Mol Psychiatry* advance online publication, 17 December 2013; doi:10.1038/mp.2013.175.
38 Huang YY, Kandel ER Recruitment of long-lasting and protein kinase A-dependent long-term potentiation in the CA1 region of hippocampus requires repeated tetanization. *Learn Mem* 1994; **1**: 74–82.
39 Schulz PE, Cook EP, Johnston D Changes in paired-pulse facilitation suggest presynaptic involvement in long-term potentiation. *J Neurosci* 1994; **14**: 5325–5337.
40 Takanaga H, Mukai H, Shimakawa M, Konishi H, Kikkawa U, Koizumi T *et al.* Functional characterization of the promoter region of the mouse protein kinase C gamma gene. *FEBS Lett* 1995; **368**: 276–278.
41 Chen KH, Widen SG, Wilson SH, Huang KP Characterization of the 5′-flanking region of the rat protein kinase C gamma gene. *J Biol Chem* 1990; **265**: 19961–19965.
42 Li L, Davie JR The role of Sp1 and Sp3 in normal and cancer cell biology. *Ann Anat* 2010; **192**: 275–283.
43 Lupien SJ, McEwen BS, Gunnar MR, Heim C Effects of stress throughout the lifespan on the brain, behaviour and cognition. *Nat Rev Neurosci* 2009; **10**: 434–445.
44 Hedges DW, Woon FL. Early-life stress and cognitive outcome. *Psychopharmacology* 2011; **214**: 121–130.
45 Roceri M, Hendriks W, Racagni G, Ellenbroek BA, Riva MA. Early maternal deprivation reduces the expression of BDNF and NMDA receptor subunits in rat hippocampus. *Mol Psychiatry* 2002; **7**: 609–616.
46 Pickering C, Gustafsson L, Cebere A, Nylander I, Liljequist S. Repeated maternal separation of male Wistar rats alters glutamate receptor expression in the hippocampus but not the prefrontal cortex. *Brain Res* 2006; **1099**: 101–108.
47 Ryan B, Musazzi L, Mallei A, Tardito D, Gruber SH, El Khoury A *et al.* Remodelling by early-life stress of NMDA receptor-dependent synaptic plasticity in a gene-environment rat model of depression. *Int J Neuropsychopharmacol* 2009; **12**: 553–559.
48 Kandel ER. The molecular biology of memory storage: a dialogue between genes and synapses. *Science* 2001; **294**: 1030–1038.
49 Lisman J, Yasuda R, Raghavachari S. Mechanisms of CaMKII action in long-term potentiation. *Nat Rev Neurosci* 2012; **13**: 169–182.
50 Li S, Tian X, Hartley DM, Feig LA. Distinct roles for Ras-guanine nucleotide-releasing factor 1 (Ras-GRF1) and Ras-GRF2 in the induction of long-term potentiation and long-term depression. *J Neurosci* 2006; **26**: 1721–1729.
51 Xu L, Anwyl R, Rowan MJ. Behavioural stress facilitates the induction of long-term depression in the hippocampus. *Nature* 1997; **387**: 497–500.
52 Yang C-H, Huang C-C, Hsu K-S. Behavioral stress enhances hippocampal CA1 long-term depression through the blockade of the glutamate uptake. *J Neurosci* 2005; **25**: 4288–4293.
53 Kim JJ, Foy MR, Thompson RF. Behavioral stress modifies hippocampal plasticity through N-methyl-D-aspartate receptor activation. *Proc Natl Acad Sci USA* 1996; **93**: 4750–4753.
54 Artola A, von Frijtag JC, Fermont PC, Gispen WH, Schrama LH, Kamal A *et al.* Long-lasting modulation of the induction of LTD and LTP in rat hippocampal CA1 by behavioural stress and environmental enrichment. *Eur J Neurosci* 2006; **23**: 261–272.
55 Rioult-Pedotti MS, Donoghue JP, Dunaevsky A. Plasticity of the synaptic modification range. *J Neurophysiol* 2007; **98**: 3688–3695.
56 Dudek SM, Bear MF. Homosynaptic long-term depression in area CA1 of hippocampus and effects of N-methyl-D-aspartate receptor blockade. *Proc Natl Acad Sci USA* 1992; **89**: 4363–4367.
57 Bienenstock EL, Cooper LN, Munro PW. Theory for the development of neuron selectivity: orientation specificity and binocular interaction in visual cortex. *J Neurosci* 1982; **2**: 32–48.
58 Danielewicz J, Hess G. Early life stress alters synaptic modification range in the rat lateral amygdala. *Behav Brain Res* 2014; **265C**: 32–37.
59 Rodgers AB, Morgan CP, Bronson SL, Revello S, Bale TL. Paternal stress exposure alters sperm microRNA content and reprograms offspring HPA stress axis regulation. *J Neurosci* 2013; **33**: 9003–9012.
60 Aisa B, Tordera R, Lasheras B, Del Río J, Ramírez MJ. Cognitive impairment associated to HPA axis hyperactivity after maternal separation in rats. *Psychoneuroendocrinology* 2007; **32**: 256–266.
61 Batalha VL, Pego JM, Fontinha BM, Costenla AR, Valadas JS, Baqi Y *et al.* Adenosine A2A receptor blockade reverts hippocampal stress-induced deficits and restores corticosterone circadian oscillation. *Mol Psychiatry* 2012; **18**: 320–331.
62 Saavedra-Rodríguez L, Feig LA. Chronic social instability induces anxiety and defective social interactions across generations. *Biol Psychiatry* 2013; **73**: 44–53.
63 Arai JA, Li S, Hartley DM, Feig LA. Transgenerational rescue of a genetic defect in long-term potentiation and memory formation by juvenile enrichment. *J Neurosci* 2009; **29**: 1496–1502.

64 Dunn GA, Morgan CP, Bale TL. Sex-specificity in transgenerational epigenetic programming. *Horm Behav* 2011; **59**: 290–295.
65 Morgan CP, Bale TL. Early prenatal stress epigenetically programs dysmasculinization in second-generation offspring via the paternal lineage. *J Neurosci* 2011; **31**: 11748–11755.
66 Dietz DM, Laplant Q, Watts EL, Hodes GE, Russo SJ, Feng J *et al*. Paternal transmission of stress-induced pathologies. *Biol Psychiatry* 2011; **70**: 408–414.
67 Guerrero-Bosagna C, Settles M, Lucker B, Skinner MK. Epigenetic transgenerational actions of vinclozolin on promoter regions of the sperm epigenome. *PLoS ONE* 2010; **5**: e13100.
68 Wolstenholme JT, Edwards M, Shetty SR, Gatewood JD, Taylor JA, Rissman EF *et al*. Gestational exposure to bisphenol a produces transgenerational changes in behaviors and gene expression. *Endocrinology* 2012; **153**: 3828–3838.
69 Dunn GA, Bale TL. Maternal high-fat diet effects on third-generation female body size via the paternal lineage. *Endocrinology* 2011; **152**: 2228–2236.
70 Carone BR, Fauquier L, Habib N, Shea JM, Hart CE, Li R *et al*. Paternally induced transgenerational environmental reprogramming of metabolic gene expression in mammals. *Cell* 2010; **143**: 1084–1096.
71 Vassoler FM, White SL, Schmidt HD, Sadri-Vakili G, Pierce RC. Epigenetic inheritance of a cocaine-resistance phenotype. *Nat Neurosci* 2013; **16**: 42–47.
72 Gapp K, von Ziegler L, Tweedie-Cullen RY, Mansuy IM. Early life epigenetic programming and transmission of stress-induced traits in mammals: how and when can environmental factors influence traits and their transgenerational inheritance? *Bioessays* 2014; **36**: 491–502.
73 Curley JP, Mashoodh R. Parent-of-origin and trans-generational germline influences on behavioral development: the interacting roles of mothers, fathers, and grandparents. *Dev Psychobiol* 2010; **52**: 312–330.
74 Gowaty PA, Anderson WW, Bluhm CK, Drickamer LC, Kim YK, Moore AJ *et al*. The hypothesis of reproductive compensation and its assumptions about mate preferences and offspring viability. *Proc Natl Acad Sci USA* 2007; **104**: 15023–15027.
75 Jirtle RL, Skinner MK. Environmental epigenomics and disease susceptibility. *Nat Rev Genet* 2007; **8**: 253–262.
76 Davies G, Tenesa A, Payton A, Yang J, Harris SE, Liewald D *et al*. Genome-wide association studies establish that human intelligence is highly heritable and polygenic. *Mol Psychiatry* 2011; **16**: 996–1005.
77 Nisbett RE, Aronson J, Blair C, Dickens W, Flynn J, Halpern DF *et al*. Intelligence: new findings and theoretical developments. *Am Psychol* 2012; **67**: 130–159.
78 Deary I, Johnson W, Houlihan LM. Genetic foundations of human intelligence. *Hum Genet* 2009; **126**: 215–232.
79 Manolio TA, Collins FS, Cox NJ, Goldstein DB, Hindorff LA, Hunter DJ *et al*. Finding the missing heritability of complex diseases. *Nature* 2009; **461**: 747–753.

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